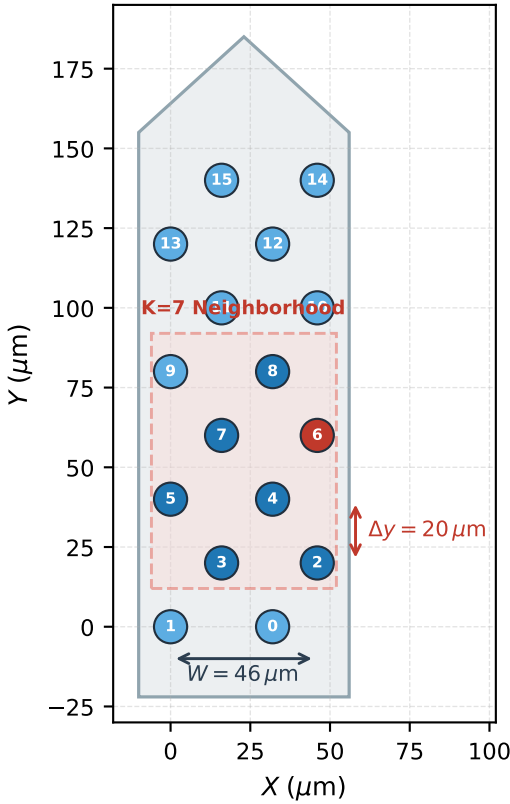


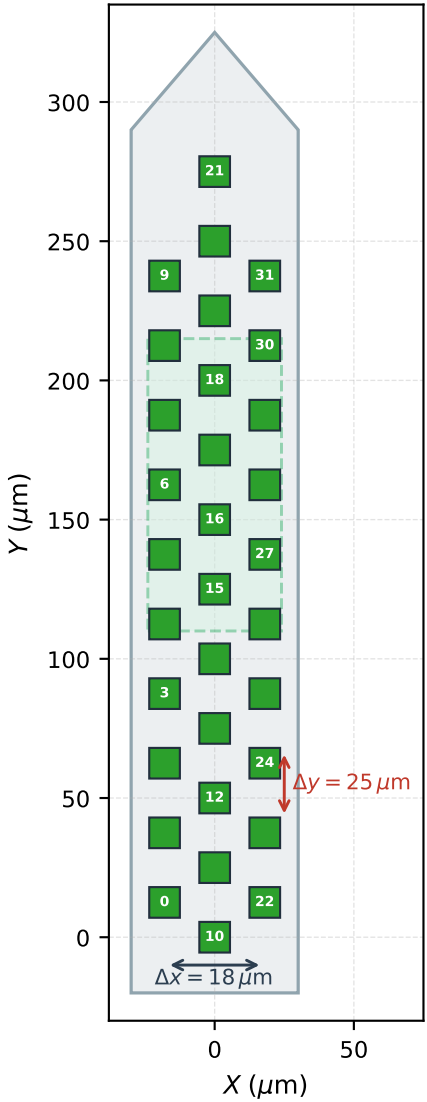
High-Density Neural Probe Architectures and Electrode Layouts Across Benchmark Datasets

(a) Hybrid Janelia
16-Ch Staggered Shank



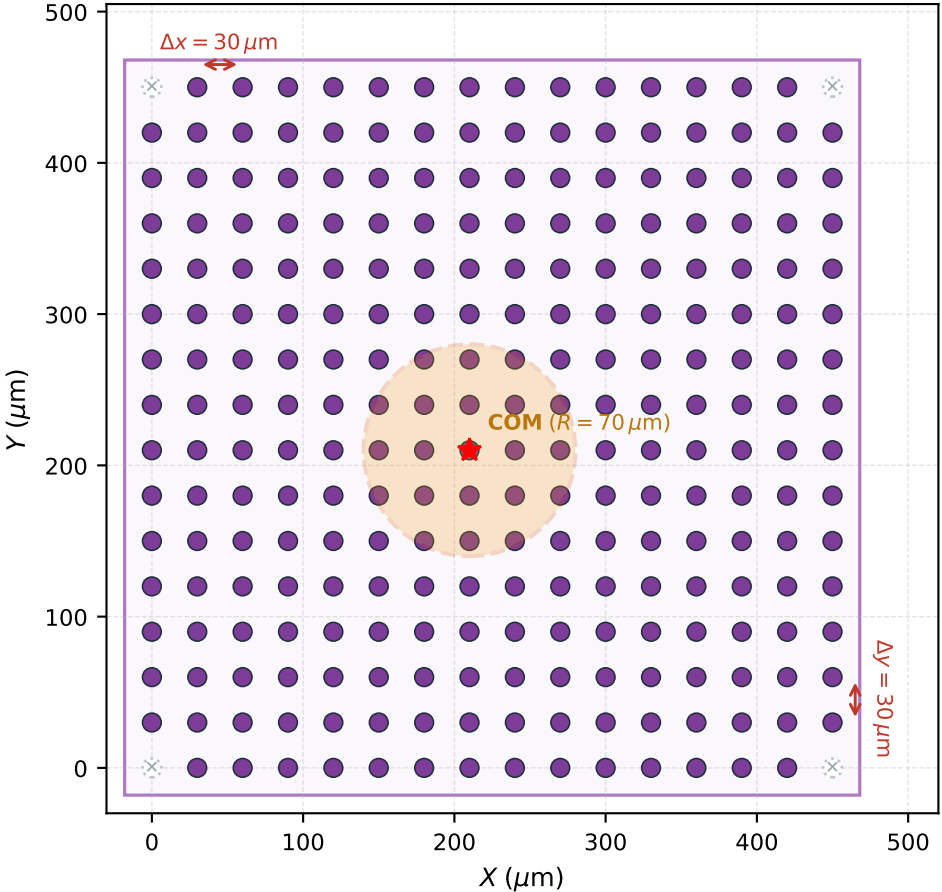
- Channels: 16
- 4 Staggered Cols
- Pitch: $\Delta y = 20 \mu\text{m}$
- Span: $46 \times 140 \mu\text{m}$
- Linear SiProbe

(b) MEArec Poly3
32-Ch Tri-Column



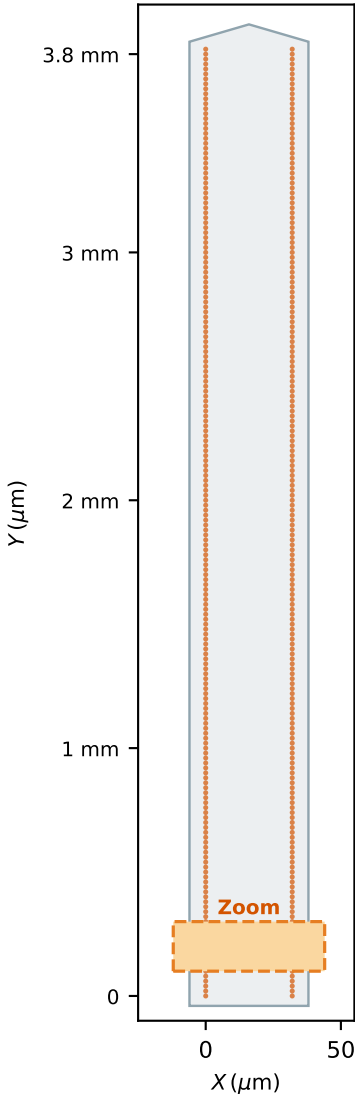
- Channels: 32
- 3-Col Hexagonal
- Pitch: $25 \times 18 \mu\text{m}$
- Span: $36 \times 275 \mu\text{m}$
- Neuronexus Poly3

(c) Yger In-Vitro Retina MEA
252-Ch 2D Planar Grid (16×16)



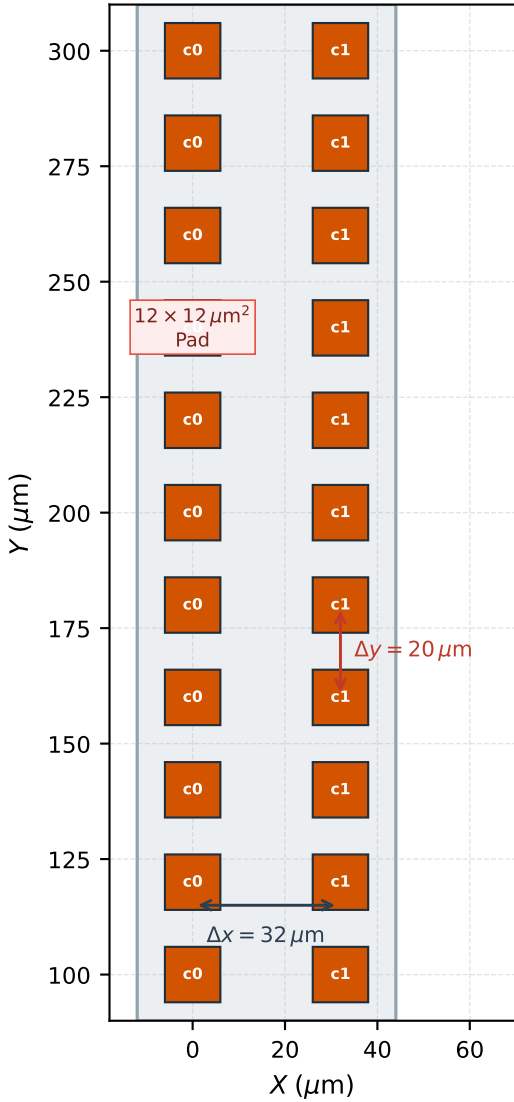
- Channels: 252
- Uniform 2D Grid
- Pitch: $\Delta x = \Delta y = 30 \mu\text{m}$
- Span: $450 \times 450 \mu\text{m}$
- Planar MEA Array

(d1) NP 1.0
Full Shank



- 384 AP Channels
- Span: 3.82 mm
- Pitch: 20 μm

(d2) NP 1.0 Detail
Checked Stagger



- Channels: 384
- Checked 2-Col
- Pitch: $20 \times 32 \mu\text{m}$
- Pad: $12 \times 12 \mu\text{m}^2$
- Penetrating Shank